

Assignment No 23

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Domain : AI & ML

College : T.B Girwalkar Polytechnic Ambajogai

Q1.

Dataset Preview:

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	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25

R2 Score: 0.575787706032451

First 10 Predictions:

Actual: 0.477

Predicted: 0.719

Actual: 0.458

Predicted: 1.764

Actual: 5.000

Predicted: 2.710

Actual: 2.186

Predicted: 2.839

Actual: 2.780

Predicted: 2.605

Actual: 1.587

Predicted: 2.012

Actual: 1.982

Predicted: 2.646

Actual: 1.575

Predicted: 2.169

Actual: 3.400

Predicted: 2.741

Actual: 4.466

Predicted: 3.916

Q2.

Dataset Preview:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness \
...					
0	17.99	10.38	122.80	1001.0	0.11840
1	20.57	17.77	132.90	1326.0	0.08474
2	19.69	21.25	130.00	1203.0	0.10960
3	11.42	20.38	77.58	386.1	0.14250
4	20.29	14.34	135.10	1297.0	0.10030
	mean compactness	mean concavity	mean concave points	mean symmetry \	
0	0.27760	0.3001	0.14710	0.2419	
1	0.07864	0.0869	0.07017	0.1812	
2	0.15990	0.1974	0.12790	0.2069	
3	0.28390	0.2414	0.10520	0.2597	
4	0.13280	0.1980	0.10430	0.1809	
	mean fractal dimension ...	worst radius	worst texture	worst perimeter \	
0	0.07871 ...	25.38	17.33	184.60	
1	0.05667 ...	24.99	23.41	158.80	
2	0.05999 ...	23.57	25.53	152.50	
3	0.09744 ...	14.91	26.50	98.87	
4	0.05883 ...	22.54	16.67	152.20	
	worst area	worst smoothness	worst compactness	worst concavity \	
0	2019.0	0.1622	0.6656	0.7119	
1	1956.0	0.1238	0.1866	0.2416	
2	1709.0	0.1444	0.4245	0.4504	
3	567.7	0.2098	0.8663	0.6869	
4	1575.0	0.1374	0.2050	0.4000	
	worst concave points	worst symmetry	worst fractal dimension		
0	0.2654	0.4601	0.11890		
1	0.1860	0.2750	0.08902		
2	0.2120	0.2612	0.09750		

```

      worst concave points worst symmetry worst fractal dimension
... 0      0.2654      0.4601      0.11890
    1      0.1860      0.2750      0.08902
    2      0.2430      0.3613      0.08758
    3      0.2575      0.6638      0.17300
    4      0.1625      0.2364      0.07678

```

[5 rows x 30 columns]

Confusion Matrix:

```

[[41  2]
 [ 1 70]]

```

Accuracy : 0.9737

Precision: 0.9722

Recall : 0.9859

F1 Score : 0.979

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.95	0.96	43
1	0.97	0.99	0.98	71
accuracy			0.97	114
macro avg	0.97	0.97	0.97	114
weighted avg	0.97	0.97	0.97	114

Q3.

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*** KNN Accuracy Results:
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k = 3 --> Accuracy = 0.9474
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k = 5 --> Accuracy = 0.9474
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k = 7 --> Accuracy = 0.9474
```

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Best k value: 3
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Best Accuracy: 0.9474
```

Q4.

✓ ... Dataset Preview:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	\
0	17.99	10.38	122.80	1001.0	0.11840	
1	20.57	17.77	132.90	1326.0	0.08474	
2	19.69	21.25	130.00	1203.0	0.10960	
3	11.42	20.38	77.58	386.1	0.14250	
4	20.29	14.34	135.10	1297.0	0.10030	

	mean compactness	mean concavity	mean concave points	mean symmetry	\
0	0.27760	0.3001	0.14710	0.2419	
1	0.07864	0.0869	0.07017	0.1812	
2	0.15990	0.1974	0.12790	0.2069	
3	0.28390	0.2414	0.10520	0.2597	
4	0.13280	0.1980	0.10430	0.1809	

	mean fractal dimension	...	worst radius	worst texture	worst perimeter	\
0	0.07871	...	25.38	17.33	184.60	
1	0.05667	...	24.99	23.41	158.80	
2	0.05999	...	23.57	25.53	152.50	
3	0.09744	...	14.91	26.50	98.87	
4	0.05883	...	22.54	16.67	152.20	

	worst area	worst smoothness	worst compactness	worst concavity	\
0	2019.0	0.1622	0.6656	0.7119	
1	1956.0	0.1238	0.1866	0.2416	
2	1709.0	0.1444	0.4245	0.4504	
3	567.7	0.2098	0.8663	0.6869	
4	1575.0	0.1374	0.2050	0.4000	

	worst concave points	worst symmetry	worst fractal dimension
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```

...   worst concave points  worst symmetry  worst fractal dimension
0      0.2654              0.4601          0.11890
1      0.1860              0.2750          0.08902
2      0.2430              0.3613          0.08758
3      0.2575              0.6638          0.17300
4      0.1625              0.2364          0.07678

```

```
[5 rows x 30 columns]
```

Confusion Matrix:

```
[[40  3]
```

```
[ 1 70]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.93	0.95	43
1	0.96	0.99	0.97	71
accuracy			0.96	114
macro avg	0.97	0.96	0.96	114
weighted avg	0.97	0.96	0.96	114

Q5.

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...

Algorithm Comparison:

	Algorithm	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.9737	0.9722	0.9859	0.9790
1	KNN (k=7)	0.9474	0.9577	0.9577	0.9577
2	Naive Bayes	0.9649	0.9589	0.9859	0.9722

Best Algorithm: Logistic Regression

Accuracy: 0.9737